

The Curse of Expertise: The Effects of Expertise and Debiasing Methods on Predictions of Novice Performance

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Experts are often called on to predict the performance of novices, but cognitive heuristics may interfere with experts' ability to capitalize on their superior knowledge in predicting novice task performance. In Study 1, experts, intermediate users, and novices predicted the time it would take novices to complete a complex task. In Study 2, expertise was experimentally manipulated. In both studies, those with more expertise were worse predictors of novice performance times and were resistant to debiasing techniques intended to reduce underestimation. Findings from these studies suggest that experts may have a cognitive handicap that leads to underestimating the difficulty novices face and that those with an intermediate level of expertise may be more accurate in predicting novices' performance.

Organizations often rely on experts to predict the performance of novices. Project managers predict team members' times to complete new work assignments. Marketers and designers predict consumers' abilities to master the use of new products. Teachers predict students' efforts to complete homework and finish exams. The predictions of experts influence the allocation of time and resources to projects, decisions about product design, and the pacing of students' lessons. Predictive errors can have significant negative consequences: employee alienation, failure to meet project deadlines, consumer dissatisfaction, and student frustration or boredom. This study draws on the literature in cognitive heuristics to explore whether experts, as compared with those with less expertise and novices themselves, are more or less accurate in predicting the perfor-

mance of novices. The measure of performance in this research was the accuracy of estimating how long it would take novices to complete a new task. Inaccuracy can result from either overestimating or underestimating how long it will take novices to complete a task. Because people generally underestimate task performance times (see Buehler, Griffin, & Ross, 1994), this study also explored the effectiveness of debiasing techniques intended to improve underestimators' predictions of novice performance.

Accuracy of Experts' Predictions

Logically, one may expect experts to be comparatively good at predicting novice performance. Experts have experience as learners and a body of task-specific knowledge on which to draw (Simon, 1989; Simon, Larkin, & McDermott, 1980). From their experience as learners, experts should be familiar with the problems a novice will encounter and should therefore be able to predict how much time it will take novices to learn. Presumably, experts can extrapolate from their task-specific knowledge and store of practical know-how to guess what difficulties novices face when performing a new task. Therefore, experts should make more accurate predictions than those with less expertise.

This research was partially supported by funding from Hewlett-Packard Laboratories. I am grateful to George Loewenstein, Sara Kiesler, Robert Kraut, and Robyn Dawes for their assistance during the formative stages of this work.

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Research on cognitive heuristics suggests a different set of predictions—that experts may not be very good at predicting novices' performance after all. Three cognitive shortcuts on which experts may rely could lead them to underestimate the performance time of novices. First, experts may rely on the availability heuristic to predict novice performance. The *availability heuristic* is a cognitive shortcut in which people make predictions on the basis of the information that they have readily available in memory (Tversky & Kahneman, 1973). Experts, if they have learned a task longer ago than those with less expertise, may not have their own learning experience as readily available in memory as those with less expertise because people tend to have better recall of their more recent experiences (Hoch, 1984; Miller & Campbell, 1959). For example, Marcus (1986) studied people's recall of their own political attitudes and found that participants' recall of their political attitudes in 1973 more closely matched their attitudes at the time of making the estimate in 1982. Experts' recent experiences, which should be more available, are likely to include a disproportionate number of trouble-free, routine experiences with the task. If the availability heuristic influences estimators' predictions of novice task performance, then experts are likely to underestimate novice task times.

A second cognitive heuristic that may bias experts is anchoring and insufficient adjustment. Experts may anchor on their own performance and fail to adjust adequately for the differences in skills between themselves and novices. People often anchor on their own attitudes, beliefs, and knowledge and use this anchor as a basis for predicting what others believe, feel, and know (Davis, Hoch, & Ragsdale, 1986; Nickerson, Baddeley, & Freeman, 1987). This prediction is consistent with *the curse of knowledge*, a bias whereby knowledgeable people are unable to ignore information they hold that others do not (Camerer, Loewenstein, & Weber, 1989). Experts have a larger gap to bridge than those with less expertise when they try to predict the skill and knowledge level of novices. From the literature on anchoring and adjusting, experts should anchor on their own skill and knowledge, which is greater than that of others and requires a larger adjustment to accurately predict the inferior skill

and knowledge of novices. Because the adjustment required is larger for experts, they should underestimate the performance of novices more than do those with less expertise.

A third cognitive shortcut that may interfere with experts' abilities to predict novice performance is oversimplification of the task. Langer and Imber (1979) argued that "the expert . . . may be in a position of knowing that he/she can perform the task, without any longer knowing *how* he or she performs it, that is, without knowing the steps or components that make up the performance" (p. 2015). As people become more expert, they automate the process (see Sternberg, 1997) and develop an oversimplified view of the task as the details of the task become less salient (Langer & Imber, 1979). In oversimplifying the task, experts may lose sight of the complexity faced by the novice performer.

If the availability heuristic, anchoring and adjusting, and oversimplification are involved when experts attempt to predict novice performance, then experts may not benefit from their superior experience and may seriously underestimate the performance of novices as compared with those with less expertise.

Novices also should underestimate task completion time because they have no task relevant experience on which to draw, and they may have unrealistically simplistic views of the task. They are likely to generate overly smooth scenarios because of their lack of task-specific experience combined with their failure to take into account the unplanned difficulties that can arise with any new project or task (see Buehler et al., 1994). Therefore, in comparison with those with a little expertise (intermediate users), novices should be less accurate estimators.

Hypothesis 1: Experts and novices will underestimate novice task completion times more than will intermediate users.

Improving Experts' Predictions

Methods for improving people's predictive and recall accuracy have been explored in numerous studies (e.g., Connolly & Dean, 1997; Glenberg, Sanocki, Epstein, & Morris, 1987; Osberg & Shrauger, 1986; Thompson & Mingay, 1991). Two such methods may improve the accuracy of

experts' predictions of novice performance by reducing reliance on cognitive heuristics. One method is to prompt people to recall their own experience and instruct them to use this experience to construct a scenario of how the task being estimated may progress. When studying the accuracy of estimating the time it would take programmers to write a new program, Buehler et al. (1994) found that research participants provided more accurate estimates of task completion times when they were required to construct scenarios linking past experience and the event to be estimated. To the extent that underestimation is caused by use of the availability heuristic, prompting experts to construct a novice scenario may make their own novice experience more available. To the extent that underestimation is caused by anchoring and adjusting, prompting experts to recall their past experiences may help them reset their anchor or make a bigger adjustment. More generally, prompting experts to recall their own experience and to develop scenarios may encourage them to process relevant information more systematically and rely less on cognitive heuristics (e.g., Tversky & Kahneman, 1974). Because novices have no task-specific experience on which to draw, asking them to reflect on their own task experience is unlikely to cause any increase in predictive accuracy.

Hypothesis 2: Experts' predictions of novice task times will increase in accuracy when they are asked to recall their own experience.

Hypothesis 3: Novices' predictions of novice task times will not increase in accuracy when they are asked to recall their own experience.

Another debiasing method that has been used in prior research (e.g., Engle & Lumpkin, 1992) is to present estimators with a list of problems novices encounter when they learn the task. Recent research indicates that this method, combined with the suggestion to remember one's own past experiences, results in significant improvements in recall accuracy. Providing a list of potential problems has significantly improved the accuracy of recall in a variety of tasks, including studying and talking on the telephone (Engle & Lumpkin, 1992). Buehler et al. (1994) used a similar debiasing method of providing information about the target's previous task experiences.

They found that time estimates increased and that observers relied heavily on the information about others' past problems to generate their predictions of the target's performance. This debiasing method may be especially effective for experts. Presenting experts with a list of novice problems might remind them of their own experiences as novices, making these experiences more available in memory. Presenting experts with a list of novice problems may also shift the experts' anchor from their own recent experience to their own experience as novices. This list also may provide experts with more information for the adjustment process as the steps in the process are made more salient. Because a list is a more explicit reminder of the novice experience than an admonition to recall one's own prior experience, one may expect the list method to be more effective at debiasing experts than the recall method.

Novices are unlikely to have anticipated the problems identified on the list and may increase their accuracy when given relevant information on the difficulty of the task. In fact, a list of problems significantly increases the relevant information available to novices in their estimation process. Therefore, the list method should increase the accuracy of novices' predictions by increasing the length of time they estimate the task will take.

Hypothesis 4: Experts will increase the accuracy of their novice task time predictions more when they are presented with a list of problems faced by novices than when they are asked to recall their own experiences.

Hypothesis 5: Novices will increase the accuracy of their novice task time predictions even more when they are presented with a list of problems faced by novices than when they are asked to recall their own experiences.

Overview of Studies

To test these hypotheses, two studies were performed. The first was a field study comparing novice, intermediate, and expert predictions of novice performance times on a series of tasks using a cellular telephone. In Study 1, three elicitation methods were compared—unaided, recall, and list. Study 2 was a laboratory experi-

ment in which expertise was manipulated using a LEGO[®] assembly task (V-Wing Fighter; Lego Systems, Inc., Enfield, CT). In this between-subjects experiment, predictions based on the unaided elicitation and the list debiasing method were compared.

Study 1

Study 1 was designed to compare people's predictions of novice performance when using an advanced cellular telephone technology. Salespeople (experts), customers (intermediate users), and novices estimated the novices' performance. This approach allowed for a measure of the accuracy of the predictions by comparing the prediction of each group with actual novice performance. Each group of research participants also was exposed to one debiasing method—the recall method (telling research participants to remember their own past experience) or the list method (presenting research participants with a list of novice problems). Both debiasing methods were modeled after approaches found successful in previous research with similar estimation tasks (e.g., Buehler et al., 1994; Engle & Lumpkin, 1992). It was expected that research participants in each group (experts, intermediate users, and novices) would tend to underestimate novices' task completion times. People generally make overly optimistic predictions, in part because they fail to take into account the number of unanticipated problems they have encountered on other similar tasks (Buehler et al., 1994; Kahneman & Tversky, 1979). Experts (e.g., Kidd, 1970), those with less expertise (e.g., Buehler et al., 1994), and novices would all be expected to underestimate task performance. Over and above this general bias toward underestimation, this study poses different predictions for groups with different levels of expertise (i.e., experts may underestimate more than others).

The design was a 3 (level of expertise) $\times$ 2 (debiasing condition) factorial in which every research participant made two predictions of novices' performance. The unaided-recall condition, assigned to half of the participants at each level of expertise, required these participants to first predict novices' task completion times on the basis of a description of the novices and the task (the unaided prediction). These same research

participants then predicted novices' task completion times after exposure to a recall debiasing method. The unaided-list condition, assigned to the other half of the participants, required that these participants make the unaided prediction first, then make their estimate after exposure to a list debiasing technique. The novices always performed the task after giving their estimates.

There were two dependent variables of interest. Predictive accuracy was calculated as the difference between the prediction and the actual (median) time it took the novices to complete the task. Using predictive accuracy allows an evaluation of the extent to which performance time was underestimated (the predicted direction of the bias).²

Method

Novice task. A task was required in which experience ranged from none to extensive. Further, it was important that experts be knowledgeable in the domain and highly skilled at the task. At the time of this study, a field trial of an advanced cellular telephone and network services (Personal Communication Service, or PCS) was underway at Carnegie Mellon University. The task given to research participants was to estimate the median length of time it would take novices to use the advanced cellular telephone to store a greeting on voice mail, leave a message on voice mail, and listen to the messages in a voice mail box, erase one message, and save one message. Novices generally had difficulty performing the task. They had to use an indicator light rather than dialtone to recognize the system was working. They had to learn to send a call using a keypad. They also had to navigate at least three levels in the voice prompt menu of the voice mail system.

Participants. Research participants were drawn from three different populations. Experts were drawn from the sales staff of a local cellular telephone company sponsoring the field trial of PCS. The sales staff in this organization is

¹LEGO[®] is a trademark of the LEGO Group (Lego Systems, Inc., Enfield, Connecticut).

²Absolute error (the absolute value of estimate accuracy) also was used in the Study 1 analyses and resulted in the same pattern of results.

responsible for selling cellular telephone service directly to customers, training customers in the use of the phones, and providing ongoing customer support. Salespeople were recruited through the company field trial coordinator. Several of the volunteers were eliminated from the study because they were recently hired by the cellular company and did not have extensive knowledge of the technology used in the trial. The 18 remaining experts fit the definition of experts in that they had extensive task related experience and had, as experienced mobile telecommunications salespeople, a large body of domain knowledge.

Research participants with an intermediate level of expertise were drawn from a sample of university staff, faculty, and students who were participating as customers in the PCS field trial. The customers in the field trial were approached in person and asked to participate in this study. Sixty-nine percent of the intermediate users were available to participate. The 44 intermediate users had 3–9 months of experience with the new cellular telephones and services.

Novice research participants were drawn from the pool of graduate students in nontechnical and nonartistic departments at the university. They were recruited at the graduate coffee house and at the graduate student lounges on campus. Novices were asked to complete estimates similar to the ones completed by the expert and intermediate research participants and to learn how to perform the tasks after completing the estimates. Thirty-four novice volunteers finished the study. Nine novices decided after volunteering that they did not have time to complete the study. A one-way analysis of variance (ANOVA) examining the difference in estimates between novices who completed the task and those who only provided estimates showed no difference in estimates between these groups.

All research participants were given a gourmet chocolate candy bar for their participation in the study.

Procedure. Novice, intermediate user, and expert research participants were asked to predict the time it would take a novice to perform a task with three components: (a) store a greeting on voice mail, (b) leave a message on voice mail, and (c) listen to the messages in their voice mail box, erase the first one (from the experimenter), and save the message they recorded in the

previous component. Research participants were given a booklet that described the novices as graduate students from nontechnical and nonartistic departments on campus and told them that the novices would be “sitting in a coffee house or in a shared office while learning to perform the tasks.” Research participants were further informed that none of the novices had ever owned a cellular phone or subscribed to voice mail and that the novices would be instructed to learn how to perform the tasks from written instructions developed by the companies providing the products.

With their estimate booklets, all research participants were given a copy of the instructional materials novices would use. (These were the actual materials used by customers in the field trial.) The experimenter asked research participants to open the booklet and make two separate time estimates. Research participants within each level of expertise were randomly assigned to presentation sets in which questions were presented in one of the following two orders: Presentation Set 1 = unaided (Trial 1)–recall (Trial 2) and Presentation Set 2 = unaided (Trial 1)–list (Trial 2). The unaided prediction simply described the situation and then read as follows:

We would like your estimate of how long you think it will take these people to learn to perform tasks A, B, and C above. Please provide a “median estimate.” This is an estimate such that you would expect half of the participants to take more time than your estimate and the other half to take less time than your estimate. Please fill in the statement below with your best estimate.

I estimate that it will take ____ minutes for a participant to learn how to perform tasks A–C above.

The above scenario is referred to as the unaided question because no additional information relevant to the task was elicited or provided in this question. The second method is referred to as the recall question and the instructions read as follows:

We would now like you to recall your own learning experience and think of some of the problems you may have encountered when you attempted to learn to use the new cellular phones and voice mail. Keeping your own experience in mind, think about any points of confusion or difficulty you think the participants might encounter in performing tasks A–C.

Research participants were further instructed to make notes of any difficulties they thought the novices would encounter, to "construct a scenario of a participant learning to perform tasks A-C," and to incorporate this information into their estimate. Space was provided for research participants to make notes about their scenario. The rest of the instructions were the same as in the unaided prediction. The final (list) question was identical to the recall question with one additional component. After listing the tasks, the instructions stated that several graduate students were observed learning the tasks and that some had difficulty. Six observed difficulties were listed, such as "figuring out how to get to a previous menu when having made an error" and "clearing the display when making a dialing error." Research participants were instructed to "construct a scenario of a participant learning to perform tasks A-C" and to use this information in developing an estimate. As in the recall question, the rest of the question mirrored the unaided prediction.

After providing the two estimates, research participants were asked to answer a series of questions regarding their experience with cellular technology and network services. They were asked when they began using a cellular phone, how many calls per day they sent and received on their cellular phone, how many voice mail messages they retrieved each day, and several questions regarding their proficiency with the voice mail system.

After experts and intermediate users finished their estimates and other questionnaire items, they were dismissed. After novices completed their estimates, the novices learned to use the cellular handset (telephone) to record, listen to, erase, and save a message on the associated voice mail service. Novices were given instructional materials that had all the necessary information to complete the task. They were required to work alone until they finished the task. Start and end times were recorded.

Analysis. The primary dependent variable was predictive accuracy—research participants' predictions minus the median actual task completion time of the novices in the study. Repeated measures ANOVAs were used to control for multiple trials and planned orthogonal contrasts (experts and novices vs. intermediate users; ex-

perts vs. novices) were made to determine the nature of the relationship between expertise and predictive accuracy. An alpha level of .05 was used for all statistical tests.

Results

The experts had used cellular phones a mean of 28.50 months ($SD = 17.72$ months) with a range of 12–72 months. The intermediate group had used cellular phones a mean of 9.32 months ($SD = 10.42$ months) with a range of 1–30 months. By selection, none of the novices had any experience using cellular telephony. The expert and intermediate groups differed in their number of months of use, $t(53) = 4.91, p < .05$, in the number of cellular calls they placed per day ($M = 8.43$ vs. $M = 5.72$), $t(52) = 1.54, ns$, and in the number of voice mail messages they received per day ($M = 6.64$ vs. $M = 2.34$), $t(52) = 4.15, p < .05$. In every case, experts were heavier users. By their own estimates, intermediate users had completed many calls (1,340 on average); however, experts had completed a much larger number (8,640 on average). It is unclear when the learning curve flattens out, but it is clear that experts had far more experience than did intermediate users. Experts, as compared with intermediates, also rated themselves as more proficient at using the voice mail system ($M = 14.23$ vs. $M = 11.02$), $t(54) = 3.32, p < .05$. In summary, the classification of experts and intermediate users successfully separated the groups by both amount of experience and perceived skill at performing the tasks. Novices, in all measures, confirmed they had no experience or task specific skill.

Accuracy. Novices, on average, took 31.5 min ($SD = 11.14$ min) to perform the task. The range was from 10 to 52 min, with a median of 33 min. Research participants, on average across conditions, predicted it would take only 18.47 min ($SD = 10.89$ min) for novices to complete the task. Both parametric and nonparametric tests comparing actual novice performance against participants' average estimates indicated a significant difference between actual and estimated performance, $t(74) = 7.42, p < .05$. These data support the considerable literature on peoples' tendencies to underestimate the time it will take

to complete future tasks (e.g., Buehler et al., 1994).

Table 1 shows the means for all groups and debiasing conditions. The first analysis compares predictions across expertise for unaided trials. Experts, on average, predicted that it would take 12.86 min ($SD = 4.23$ min) for a novice to perform the task, intermediate users predicted it would take 20.09 min ($SD = 11.97$ min) on average, and novices predicted it would take 15.71 min ($SD = 9.46$ min) on average. Differences in unaided predictions by people of different expertise were tested using a one-factor ANOVA with three levels of expertise (novice, intermediate user, and expert). The ANOVA results indicated a significant difference in prediction by level of expertise, $F(2, 92) = 3.83, p < .05$. To test Hypothesis 1, a planned contrast was performed comparing experts and novices with intermediate users. Hypothesis 1, that experts and novices would be more biased toward underestimation, was supported, $F(1, 93) = 7.50, p < .05$. To further explore the relationship between expertise and underestimation, a second orthogonal contrast comparing experts with novices was performed. The results indicated no significant difference between the predictions of experts and novices, $F(1, 93) = 0.95, ns$.

Debiasing. The debiasing hypotheses posited that experts would increase the accuracy of their predictions of novice performance when asked to recall their own experience (Hypothesis 2) and when presented with a list of problems previously encountered by novices (Hypothesis 4) but particularly when presented with the list of problems (Hypothesis 4). The data from Table 1 suggest that intermediate users (and novices) improved but experts did not when given ques-

tions designed to reduce the biases toward underestimation. Figure 1 illustrates the improvement from unaided to debiased predictions by group. To directly test Hypothesis 2, a repeated measures ANOVA was performed, including experts' predictions in only the unbiased-recall condition, with trial as the within-subjects variable. The results indicated no significant improvement from the unaided to the recall debiasing method, $F(1, 9) = 0.03, ns$. A similar analysis was performed with experts in only the unaided-list condition to test Hypothesis 4. Again, the results showed no significant improvement from the unaided to the list debiasing method, $F(1, 7) = 0.10, ns$. Hypothesis 4 posited that the list debiasing method would be more effective at reducing experts' bias toward underestimation than the recall method. To test this element of the hypothesis, a repeated measures ANOVA was performed. The results suggested no difference between experts' improvement on the recall versus the list debiasing methods, providing no support for Hypothesis 4.

To test Hypothesis 3, that novices will not improve when presented with the recall method, a repeated measures ANOVA was performed, including novices in only the unaided-recall condition, with trial as the within-subjects variable. The results indicated a marginally significant decrease from the unaided to the recall method, $F(1, 17) = 3.10, ns$. A similar analysis was performed to test Hypothesis 5, that novices will improve when presented with the list method, with novices in only the unaided-list condition. Contrary to Hypothesis 5, the results showed a very large, although not statistically significant, improvement from the unaided to the list debiasing method, $F(1, 15) = 2.35, ns$.

Table 1
Mean Estimates and Standard Deviations (in Minutes) as a Function of Debiasing Condition, Trial, and Level of Expertise

Level of expertise	Unaided-recall ($n = 48$)				Unaided-list ($n = 47$)			
	Trial 1		Trial 2		Trial 1		Trial 2	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Novice	13.39	5.15	14.86	6.16	18.34	12.37	20.84	13.36
Intermediate	19.38	9.31	21.63	10.33	20.72	14.06	26.35	16.71
Expert	12.95	4.92	12.75	6.75	12.75	3.51	12.31	4.30

Note. Actual median novice time was 33 min; therefore, the higher the mean in the table, the closer the estimate of the median was to the actual median performance.

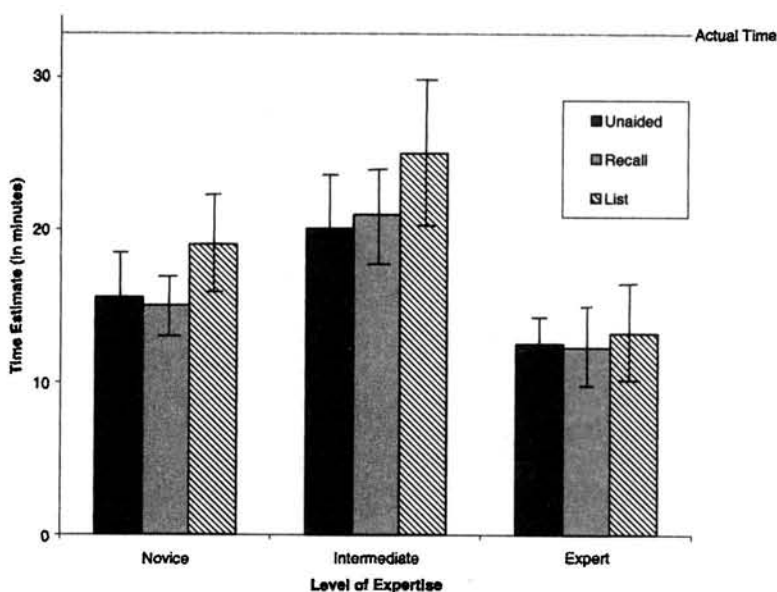


Figure 1. Estimation accuracy in Study 1 by level of expertise and debiasing condition. Error bars represent 95% confidence intervals.

To further explore the relationship between expertise and debiasing, both debiasing conditions were used in a repeated measures analysis in which the repeated measures were Trial 1 and Trial 2, and the between-subjects variables were the group (expert and intermediate users only) and the debiasing condition. In both debiasing conditions combined, intermediate users improved their predictions by 20%, whereas experts gave predictions that were 2% worse in the debiased than in the unaided trials. The results of this analysis indicated no main effect for trial, $F(1, 57) = 3.19, ns$, and a significant interaction between trial and level of expertise, $F(1, 57) = 4.41, p < .05$, suggesting that experts were more resistant to debiasing, in general, than intermediate users.

Discussion

Consistent with Hypothesis 1, the experts in this study underestimated novice performance times. In fact, they underestimated significantly more than did intermediates. Hypotheses 2 and 4 predicted that experts' predictions would be more accurate when experts were asked to recall their own experience and when provided a list of problems, respectively. Neither of these hypoth-

eses were supported by the data. In fact, both debiasing conditions when compared with unaided trials resulted in less accurate predictions by experts. In the list condition, experts were significantly less responsive to debiasing than were intermediate users.

As expected, novices were also poor predictors of novice performance times. Although novice estimators were more accurate than experts, they were less accurate than intermediate users. These results are not surprising. Novices, in comparison with intermediate users, should have little understanding of the task they are being asked to predict and are unlikely to understand the subtasks involved. In the absence of such information, estimators have a tendency to believe that tasks will go along more smoothly than is realistic, leading to this general bias toward underestimation (Buehler et al., 1994; Kahneman & Tversky, 1979). Novices improved their predictions in both debiasing conditions. This result was not expected. Novices were not expected to improve their predictions when prompted to recall their own past experience with the task because they had no task specific experience. There are at least two possible explanations for the results. First, novices might have improved in the recall presentation simply because they were asked to estimate

again. Having little confidence in their first estimate, they might have assumed that they were missing something when asked to provide a second estimate. Second, the recall debiasing might have caused novices to think more systematically about the task and even begin to consider the fact that there are often unforeseen events that extend the time it takes to learn any new task. Previous research has established that people have difficulty using relevant past experience in predicting future events (e.g., Kahneman & Tversky, 1979). When prompted to think about and develop a scenario that incorporates the past, novices might have used similar experiences in lieu of specific task experience. Such an explanation is consistent with the findings of Buehler et al. (1994) who found that novice estimators improved their estimates when asked to develop a scenario using past experience.

There are several limitations to this study. First, in order to find experts, intermediate users, and novices, three different samples were identified. These were a group of sales staff, recent customers, and novices. An alternative explanation for the results may be that the different groups had different motivations for producing either high or low estimates. For instance, novices may have been motivated to underestimate the task difficulty because they knew they were going to be performing the task. They may have been hoping that the task would be quick so that they could finish the task and get back to their own activities. Experts in this study were salespeople at a cellular company. They may have been motivated to underestimate task difficulty because they wanted to believe that cellular products are easy for people to learn to use. As a salesperson, it may even be functional to retain a bias toward underestimation because ease-of-use is one basis on which many products are sold. Such thinking can make the desired event or scenario seem more probable (Weinstein, 1980). If salespeople underestimate the difficulty of learning to use their product even when told of novice problems, their abilities to improve product design and respond to customer needs may be hindered.

Second, participants in Study 1 were all given the unaided elicitation followed by either the recall or the list debiasing method. This within-

subjects design may have resulted in a demand effect whereby participants increased their second estimate simply because they were asked to make a second estimate. It also is possible that experts were more confident in their estimate, thus less susceptible to the demand effect.

Third, Study 1 demonstrates experts' estimations of novices performance in one task domain. Replication is necessary to generalize the results to other task domains. Evidence suggests that some tasks are prone to underestimation whereas others are not (Byram, 1996). Determining the pervasiveness of experts' inaccuracy in predicting novice performance thus requires a representative sampling of tasks.

Study 2

Novices, intermediates, and experts were drawn from different samples in Study 1, and the difference between their predictions can be explained by differences in those populations. To eliminate this alternative explanation of the Study 1 findings, expertise was manipulated in Study 2. Further, one possible explanation for the effectiveness of the debiasing methods in Study 1 was a demand effect caused by asking participants to make a second estimate. To avoid the ambiguity caused by a possible demand effect, Study 2 was a between-subjects design in which participants were given either an unaided or a debiasing elicitation. Finally, a different task domain was selected to ensure generalizability beyond the cellular telephone task.

Study 2 was a 2×2 between-subjects design with expertise (low, high) and debiasing (unaided, list) as variables. Participants were asked to build a LEGO toy and to predict how long it would take the average student to build the same toy the first time. The LEGO toy was a 39-piece V-Wing Fighter (see Figure 2). Participants were given two pictures of the assembled toy—one side view and one front view—but were not given step-by-step instructions for assembling the toy. In pretests, novices took about 15 min to build the toy the first time. Expertise was manipulated by asking those in the high expertise condition to practice building the LEGO toy five times over two sessions that spanned several

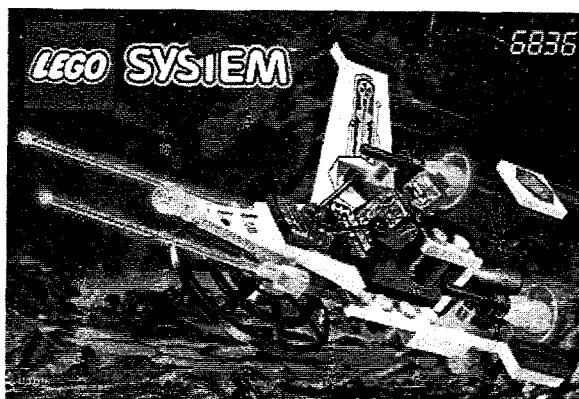


Figure 2. Illustration of LEGO V-Wing Fighter assembled by participants in Study 2. ©1997, The LEGO Group. Used with permission.

weeks. Those in the low expertise condition built the LEGO toy only once.

Method

Participants. Participants were 49 male undergraduate students in nontechnical disciplines (e.g., business, art, economics, psychology, and so forth) and were recruited through electronic and physical bulletin boards on campus. Participants were paid \$15 for their participation.

Expertise manipulation. The development of expertise relies on extensive practice and usually develops over time (Chase & Ericsson, 1982; Chase & Simon, 1973; Ericsson & Charness, 1997). In this study, participants attended two sessions. In the first session, those in the high expertise condition completed a pretest survey and assembled the LEGO toy four times before the experimenter dismissed them. Participants in the low expertise condition completed the pretest survey and were dismissed without building the LEGO toy. The second session was scheduled 10 to 14 days after the first session. In the second session, all participants built the LEGO toy one time and were then asked to provide a time estimate of how long it would take the median novice to complete the task. Therefore, when making their prediction of novices' performance times, those in the high expertise condition had built the toy five times and those in the low expertise condition had built the toy only once.

Debiasing manipulation. Participants were asked to make their predictions on the basis of

either the unaided or the list elicitation. In all conditions, participants were told the following:

We are planning a future experiment and need an estimate of how long it will take people to construct this same toy for the first time. These people will be undergraduate students at Carnegie Mellon University in nontechnical majors (i.e., industrial management, literature, etc.) and will be constructing the toy in an environment similar to this one. They will be given instructions and pictures from which to work that are identical to those given to you.

The list debiasing method was selected because it was the most powerful of the two debiasing methods used in Study 1. Similar to Study 1, the unaided elicitation simply asked participants to provide an estimate of "the median time it would take for a person like the one described above to build this toy for the first time." The list debiasing method provided a list of activities required to build the toy (i.e., figuring out how to connect the LEGO pieces) and problems novices encountered in the pretest (i.e., differentiating pieces that look the same). Participants in the list debiasing conditions were asked to "write out a scenario of what a person's experience might be like, given the potential problems listed above" and then to give a median estimate in minutes.

Procedure. All participants attended two sessions between 10 and 14 days apart. In Session 1, participants arrived at the laboratory and were asked to complete a pretest that included demographic questions and questions about their experience with LEGO toys. If they were randomly assigned to the low-expertise condition, participants completed the pretest, confirmed their second appointment, were paid \$5 for their participation at the first session, and were dismissed. Participants randomly assigned to the high-expertise condition completed the pretest and were then instructed to build the toy. When they were finished building the toy, the experimenter verified that the toy had been built correctly, gave the participants another unassembled V-Wing Fighter, and instructed the participants to build the toy again. After the participants built the toy four times, the experimenter confirmed their second appointment, paid the participants each \$5 for the first session, and dismissed them.

In the second session, participants arrived in the same laboratory, and all were instructed to build the toy. After the participants built the toy once, they were asked to provide an estimate of the median time it would take a novice to complete the task under similar conditions. They then completed a posttest survey that included questions about the participants' confidence in their estimate, the process they used to generate their estimate, the steps in the LEGO task, and their own performance time(s). After completing the posttest, participants were debriefed, paid \$10 for their participation in the second session, and dismissed.

Analysis. The primary dependent variable was predictive accuracy—participants' estimates minus the median actual task completion time of novices in the study. Median novice task completion time was calculated on the basis of the first performance time of participants in both the low and the high expertise conditions. ANOVAs were used to determine the relationship between expertise, debiasing method, and predictive accuracy.

Results

When completing the task the first time, participants on average took 12.5 min ($SD = 5.08$ min) to complete the task. Novice task performance times ranged from 3 to 27 min, with a median of 12.27 min. Research participants, on average across conditions, predicted it would take only 10.7 min ($SD = 5.68$ min) for novices to complete the task. These data are consistent with Study 1 and with other research identifying a general bias toward underestimation. When performing the task for the first time, there was no significant difference in the performance times of those in the high-expertise versus the low-expertise conditions, $F(1, 46) = 0.09$, ns , with those in the high-expertise condition taking 12.7 min ($SD = 5.77$ min), on average, and those in the low-expertise condition taking 12.3 min ($SD = 4.40$ min), on average.

Table 2 shows the mean estimates for all conditions. Those in the high-expertise condition, on average, predicted that it would take 8.5 min ($SD = 3.36$ min) for a novice to perform the task. In contrast, those in the low-expertise condition predicted that it would take 12.9 min ($SD = 6.73$

Table 2
Mean Estimates and Standard Deviations (in Minutes) by Debiasing Condition and Level of Expertise

Level of expertise	Unaided ($n = 26$)		List ($n = 23$)	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Low	11.46	5.44	14.64	7.92
High	8.31	2.29	8.75	4.33

Note. Actual median novice time was 12.27 min.

min) for a novice to perform the task. Whereas those high in expertise underestimated novice performance time by nearly 4 min, those low in expertise overestimated novice performance time by about 35 s.

Study 2 was designed as an additional test of Hypotheses 1 and 4. One may recall that Hypothesis 1 posited that experts would be more biased toward underestimation than would be those with less expertise. To test Hypotheses 1 and 4, a two-factor ANOVA was performed with expertise (low and high) and debiasing condition (unaided and list) predicting accuracy. Accuracy was calculated by subtracting participants' estimates from the actual median performance time of novices in this study. The ANOVA results indicated a significant difference in prediction by level of expertise, $F(1, 45) = 8.96$, $p < .05$. A Welch-Satterwaite t test comparing the estimates of novices and experts confirms a difference in these distributions, $t(33) = 2.88$, $p < .05$. As in Study 1, those with more expertise more severely underestimated novice performance times.

The data from Table 2 suggest that people low in expertise improved when given the list debiasing method but that people high in expertise were again resistant to debiasing. Although those in the low-expertise condition increased their estimates by over 3 min when given the debiasing list, those in the high-expertise condition increased their estimate by less than 30 s. The ANOVA results indicated no main effect for debiasing, $F(1, 45) = 1.44$, ns , and no interaction effect between expertise and debiasing, $F(1, 45) = 0.82$, ns , suggesting that neither group was particularly responsive to the list debiasing method. Because they were so close in their estimate without debiasing, the list reduced predictive accuracy of those low in expertise. Those in the unaided condition underestimated less than 1 min

(at 11.5 min), whereas those in the list debiasing condition overestimated by over 2.5 min (at 14.6 min).

To explore the mechanisms contributing to experts' bias toward underestimation, participants in Study 2 were asked about their perceptions of the task and about the process they used to generate their estimate. To determine the extent to which experts' oversimplification of the task contributed to underestimation, participants were asked to identify the steps involved in building the LEGO toy. These steps were counted, and a variable was created to represent the number of steps listed by each participant. To test the difference in number of perceived steps by level of expertise, an ANOVA was conducted with expertise and debiasing condition predicting number of steps. The results indicated no significant difference between those high and those low in expertise in the number of steps identified in the task, $F(1, 47) = 1.35$, *ns*. In fact, those high in expertise listed, on average, 6.0 steps ($SD = 4.31$ steps), as compared with those low in expertise who listed 4.9 steps ($SD = 2.00$ steps). These data suggest that oversimplifying the task by reducing the task to fewer steps was not a significant contributor to experts' underestimation. Although not statistically significant, there is some evidence that those high in expertise recalled more steps when given the list debiasing elicitation than did those low in expertise. Those in the high expertise condition listed 4.9 steps in the unaided and 7.2 steps in the list condition (over two additional steps), whereas those low in expertise listed 5.0 steps in the unaided condition and 4.7 in the list condition. These data, combined with the predictions of novices' performance, suggest that merely being aware of the steps did not assist experts in improving their estimates.

To understand the contribution of anchoring and insufficient adjustment to experts' underestimation, several additional analyses were conducted. In the posttest, participants were asked to "explain how you went about making the above estimate" and to "describe your thought process in as much detail as possible." These data were coded to reflect whether or not participants explicitly identified an anchor and whether or not participants explicitly identified an adjustment

process. For example, 1 participant described the following process:

I sat for a minute and tried to remember, in as much detail as possible, how long it took me the first time. Then, I took into account what I remembered to be the more challenging sections of the building process and figured people might need a little more time than I did.

This participant selected his own first performance as an anchor and tried to adjust for people who needed a little more time. In fact, over 80% of the participants in Study 2 stated that they had used their own performance as a point of reference for their prediction of others' performance. Over 60% of the participants also described some type of adjustment process. This evidence clearly suggests that participants used (or believed they used) their own performance as an anchor and, in most cases, adjusted from that anchor. Given similar approaches to generating the prediction, how can we explain the difference in accuracy between those high and those low in expertise?

An analysis of participants' estimates of their own performance may address this question. Two new variables were created to analyze the accuracy of participants' estimates of their own performance. A measure of participants' accuracy in recalling their own first performance was created by subtracting their estimate of their own first performance time from their actual first performance time (recorded by the experimenter). A similar measure was created as a measure of participants' accuracy in recalling their most recent performance. An ANOVA with expertise predicting the accuracy of participants' estimates of their own first performance revealed that those in the high-expertise conditions were significantly less accurate in recalling their own first performance time than were those in the low-expertise conditions, $F(1, 46) = 4.60$, $p < .05$. Those high in expertise underestimated their own first performance by over 3 min, whereas those low in expertise underestimated by less than 30 s. When asked about their most recent performance, however, those high in expertise were not significantly less accurate than those low in expertise, $F(1, 46) = 1.25$, *ns*. An additional analysis explored the mediating effect of participants' memory bias on estimation accuracy. The results indicated a relationship between expertise and

recall accuracy (own first performance), $t(46) = 2.15$, $p < .05$, a relationship between expertise and estimation accuracy, $t(47) = 2.91$, $p < .05$, and a relationship between recall accuracy and estimation accuracy, $t(46) = 3.03$, $p < .05$. When regressing estimation accuracy on both expertise and recall accuracy, the relationship between expertise and estimation accuracy diminishes in power (from $p = .005$ to $p = .041$). This suggests that experts' inability to recall their own novice performance mediates the relationship between expertise and estimation accuracy. Taken together, these analyses point to an availability bias. That is, experts severely underestimate novices' performance because they are attempting to use their own performance as an anchor, but they are unable to accurately recall their own novice performance.

Discussion

The results of Study 2 confirm experts' underestimation of novice performance times. As in Study 1, those with more expertise more severely underestimated the task performance times of novices and were resistant to debiasing. Although those in the high-expertise conditions predicted longer performance times when prompted with the list, the difference between the unaided and the list methods was minimal ($M = 8.31$ vs. $M = 8.75$). Those with less expertise increased their prediction more in the list debiasing condition ($M = 11.46$ vs. $M = 14.64$), although this difference was not statistically significant. These results suggest resistance to the list debiasing method at both low and high levels of expertise.

A closer examination of participants' prediction processes in Study 2 provides some evidence that experts attempt to anchor on their own experience as novices but are unable to accurately recall their own novice experience. This is consistent with the availability bias and suggests that experts' more recent, trouble-free performance of the task is guiding their prediction processes.

Although Study 2 addresses many of the limitations identified in Study 1, one additional limitation should be addressed. True expertise is a condition that develops over time with extensive practice and may be difficult to create in a laboratory experiment. Some have argued that

100 h or more of practice over 10+ years are required to become expert in cognitive tasks (Anderson, 1982; Ericsson & Charness, 1994). As Sternberg (1997) has argued, however, people are not expert or nonexpert but reside somewhere along a continuum of expertise. In Study 2, a relatively simple task was selected so that some degree of expertise could be acquired quickly. Participants in the high expertise condition performed the task a total of five times. These participants may not be experts in the pure sense of the term, but they have acquired enough expertise to outperform novices and to have reached a plateau of performance. Experts, on average, took less than 6 min to build the LEGO toy the fifth time they built it compared with nearly 13 min the first time. Also, their performance began to plateau at the third build when they completed it in 3.95 min ($SD = 1.05$ min), on average. Arguably, Study 2 did not test real experts, but as a compliment to Study 1 in which real experts were used, Study 2 provides support for the finding that expertise can increase the bias toward underestimation and can be resistant to debiasing.

General Discussion

Although it may be expected that experts' superior knowledge and experience should lead them to be better predictors of novice task completion times compared with those with less expertise, the findings in this study suggest otherwise. The results reported here suggest that experts' superior knowledge actually interferes with their ability to predict novice task performance times. Not only were experts unable to take advantage of their knowledge and experience, but they were also unable to correct their estimates when they were prompted with a presentation intended to help them reduce their underestimation. The inaccuracy of experts and their resistance to prompting has a number of possible explanations. Three biasing processes were explored in Study 2: oversimplification, availability, and anchoring and insufficient adjustment.

In contrast to previous research (e.g., Langer & Imber, 1979), evidence from Study 2 suggests that experts did not perceive fewer steps in the task than did those with less expertise. Therefore, oversimplification is unlikely to be a major

contributor to experts' bias toward underestimation. Evidence does suggest heavy reliance on an anchoring and adjustment heuristic with experts' own novice performance as the reference point. However, those at all levels of expertise described a similar anchoring strategy when asked to describe their estimation process. The primary difference between those with more and those with less expertise was the accuracy with which they recalled their own novice performance of the task. Experts underestimated their own novice performance times by over 3 min. They underestimated other novices' performance times by nearly 4 min. Experts' bias toward underestimating their own novice performance, combined with an anchoring process that relies on accurately recalling their novice performance (or making an adequate adjustment), nearly guaranteed experts' underestimation of others' novice performance times. Evidence from these data suggest that the availability bias is the key contributor to experts' relative inaccuracy in estimating novice performance times.

Experts in both studies were also resistant to debiasing. There are several possible explanations for the experts' resistance. First, experts might have had difficulty understanding the challenges faced by novices, even when reminded of these challenges. Even when provided a list of problems, experts might have been unable to understand novices' difficulties in identifying the problem, finding a solution, and implementing that solution. In using the cellular handsets, many novices attempted to initiate a phone call prior to the handset being turned on and connected to the network. Novices would send a call prior to the indicator light turning green and receive a high pitched error tone. Not knowing what the tone meant, the novices often attempted to send the call again or turned the handset off and on again. Both strategies usually resulted in a repetition of the error tone. Many novices were not able to figure out why they were getting the error tone and became frustrated. Experts, even when reminded that novices may not know to wait for the indicator light, did not see this as a challenging problem that could be time consuming to resolve. Such an explanation is consistent with the curse of knowledge, a bias in which knowledgeable people are unable to ignore their own superior knowledge in trying to make naive predictions

(Camerer et al., 1989). It is also consistent with research suggesting that people underestimate the impact of situational factors on others' performance (e.g., L.D. Ross, Amabile, & Steinmetz, 1977).

Second, experts might have been unduly confident in their original estimates. Experts have been found to be overconfident in judgments related to their field of expertise and to become increasingly confident, although not more accurate, with increasing information (Oskamp, 1965). In fact, less than 15% of the experts in Oskamp's study changed their answers when given additional pertinent information. Similarly, only 22% of the experts in Study 1 increased their estimates when prompted. Experts might have been overconfident in their original predictions of novice task performance times and been unwilling to reconsider their estimates when presented with the recall and the list methods. However, data from Study 2 do not support this line of reasoning. When asked about their level of confidence in their estimate, those high in expertise reported marginally lower levels of confidence than those low in expertise, $F(1, 47) = 3.31, p < .10$. Therefore, overconfidence alone is an unlikely explanation for experts' resistance to debiasing in these studies.

Third, there are a multitude of possible debiasing tools that could be developed. In the research reported here, I examine only two. It is possible that there are other debiasing methods that could improve experts' accuracy. The results of Study 2 suggest that experts engage in an anchoring and adjustment process in which their anchor is biased by the unavailability of their own novice experience. This finding suggests several possible directions for debiasing tools. One direction is to prompt experts to use a different anchor, possibly suggesting that they think about a task they have recently learned or watched someone else learn. Another way to reset the anchor is to prompt experts to think about the possible range of completion times. In a recent study, Connolly and Dean (1997) found that students with some programming background significantly improved the accuracy of their self-predictions when trained to consider the lowest and highest plausible performance times for the task. Similarly, M. Ross and Newby-Clark (1998) reported that self-predictions were more realistic when people

were prompted to generate a worst case scenario. Future research should be conducted on prompting experts to develop worst case scenarios and think about the plausible range of novice completion times. Another approach to debiasing may be to increase people's motivation for accuracy. For instance, increasing people's accountability for their judgments can significantly reduce cognitive biases (Tetlock, 1992).

Although future research on debiasing tools is important, it may be that the expert bias runs too deep and will remain resistant to debiasing methods. If that is the case, there may still be several alternatives for improving experts' estimates of novice performance times. One obvious alternative is to build expertise in predicting novice performance. As experts receive feedback on the accuracy of their estimates, they are likely to improve. Some organizations build such expertise in management recruiters, marketeers, usability engineers, and other professionals. Another alternative is to use intermediates to assist in predicting novice performance. I would not suggest eliminating the expert from this process but closing the gap by adding in the perspective of someone closer to the experience of the novices being predicted. Also, having experts observe novices may help experts to understand the problems encountered by novices. This is a common practice in human-factors laboratories, although anecdotal evidence of its effectiveness at reducing experts' bias is mixed. Finally, gathering data directly from novices on their ability or task completion times is a method of bypassing the biases involved in relying on others' judgment and could result in the most accurate estimates of novice task completion times.

There are several limitations to these studies. First, researchers have suggested that an expert's performance is sometimes tested with tasks outside the domain of the expert's experience (e.g., Bolger & Wright, 1992). One could argue on the one hand that the participants in these studies were not expert at estimating task completion times of novices and that given adequate experience in estimating novices' performance, they would become more accurate. On the other hand, the salespeople (experts) in Study 1 had extensive experience working with novices (potential and new customers). They had substantial experience in helping customers learn to use cellular technol-

ogy. The present results suggest these experiences were not very helpful in creating experts at estimation. Future research could explore how experts' experiences in estimating others affects their own accuracy. Research on experts' abilities to learn from different kinds of feedback would also be valuable. Another limitation is that only two tasks were examined in these studies, and both tasks required that participants interact with a physical object(s). It is possible that experts may respond differently with more cognitively demanding and more cognitively complex tasks. Replication with tasks in other domains and with more complexity are necessary to determine the generalizeability of these results. Third, although the expert participants in Study 1 were relatively expert in cellular telephony, they would not have been internationally renowned as experts in the domain. It is possible that exceptionally talented people within a domain are less likely to have this bias. Future research with participants located elsewhere on the continuum of expertise is warranted to fully understand the shape of the relationship between expertise and predictive accuracy.

The research reported here addresses experts' predictions of novices' future performance and not the relationship between expertise and evaluations of current performance. Extensive research has been reported on the relationship between cognitive operations and the accuracy of evaluations of observed performance (e.g., DeNisi, Cafferty, & Meglino, 1984; Feldman, 1981; Ilgen & Feldman, 1983). This work has been concerned primarily with how people encode, store, and retrieve information about observed performance rather than with how people predict future performance (for an exception, see Zalesny & Highhouse, 1992). In the study reported here, the research participants gave their predictions without observing novices' task performance. Therefore, my data cannot directly address biases in performance evaluation. However, if experts think novices' tasks will be easy and are unable to adjust that perception based on new information about the difficulties faced by novices, it follows that experts also may be particularly harsh evaluators of novices' performance.

These findings have implications in a variety of settings in which people are predicting the performance of those new to an organization,

product, or task. The results suggest that project managers may not be the most accurate at estimating project completion times for novice team members. Marketers and designers may not be the best estimators of what novices will experience when learning to use a new product. Also, teachers may not be the best estimators of how long it should take students to learn a new topic or skill. Further, these data suggest that urging experts to improve their predictions of novice performance may not be productive. If experts have a cognitive handicap when it comes to predicting novices' experiences, then applying more pressure may only cause frustration. Identifying effective debiasing methods or structuring the design process so that experts' bias is balanced with a less biased perspective, perhaps an intermediate, may result in more accurate time estimates and more usable products.

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Received January 23, 1997

Revision received November 12, 1998

Accepted December 3, 1998 ■